

Emergency Green Corridor

Warning the junction constable before the siren arrives.

Nandish · [green-corridor-2026.web.app](#)

60 %+

of Karnataka 108 cardiac, stroke, respiratory calls missed the 10-minute target

Source: CAG audit of Karnataka 108 EMS, 2014–19

Team details

A solo entry, built end to end by one contributor.

TEAM NAME

Green Corridor

TEAM LEADER

Nandish

CONTRIBUTORS

Sole contributor

PROBLEM STATEMENT

Sustainability & Social Impact: emergency vehicles stuck at red signals while the junction constable is never told.

Brief about the idea

Warn the junction constable early, sized to the queue

≈ 7.9 min

saved for the critical ambulance

4 vs 1 min

warning at a 400 m queue vs a
100 m queue

16.5 min

three-vehicle total, simulated
baseline

Scripted scenario and synthetic voices, not field results.

Opportunities (how it differs / how it solves / USP)

Not a fixed radius, and not waiting for new hardware



HOW IT DIFFERS

**Warning sized to
the queue**

vs fixed-radius alerts



HOW IT SOLVES

**Queue cleared
before the siren**

vs manual green corridors



USP

**Rules decide,
Gemini explains;
no hardware**

vs IR/GPS preemption

List of features

Six screens and the engines behind them

Suggested tier

CRITICAL

Crew confirms

Confirm CRITICAL

Voice triage

STOP

ESCALATED

Stop cross traffic immediately.

Spoken cop alert

Vitals

SBP

85

SBP 86 → 85

HR

112

SpO2

90

SpO2 89 → 90

Treatment so far: Aspirin 300 mg

Hospital live brief

Control room board

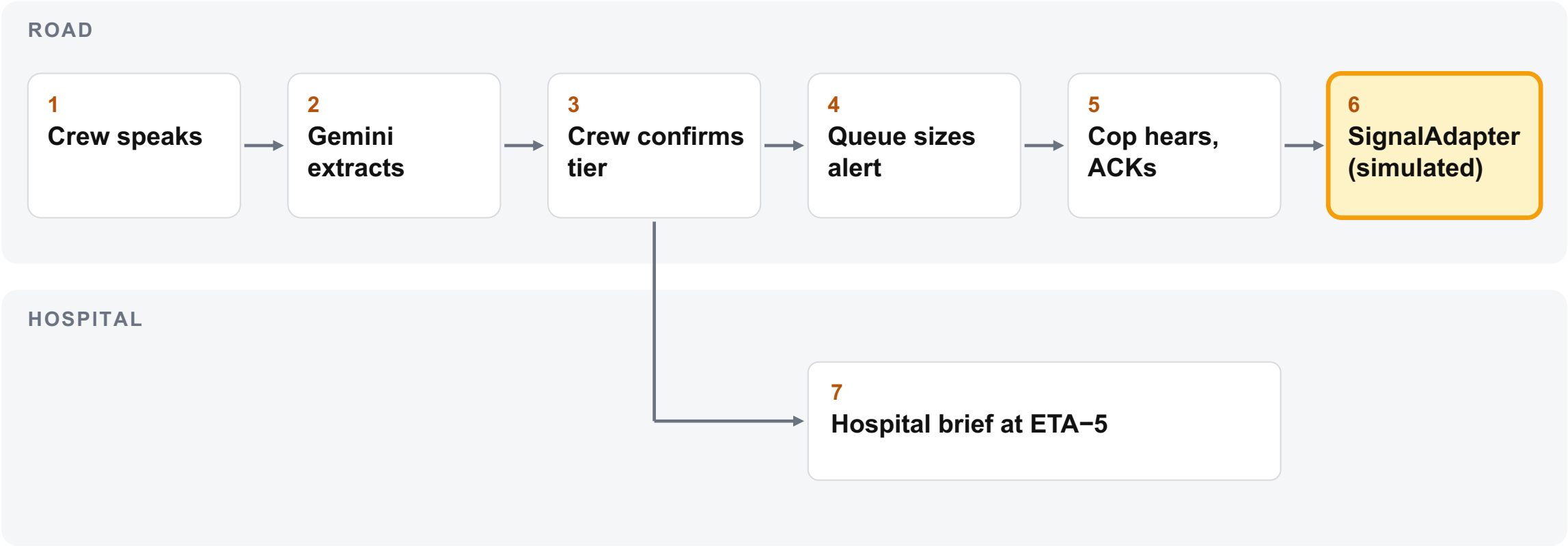
With-vs-without replay

Routing agent trace

- Lead time
- Sequencing
- Cop voice
- ATMIST
- After-action
- Eval harness

Process flow

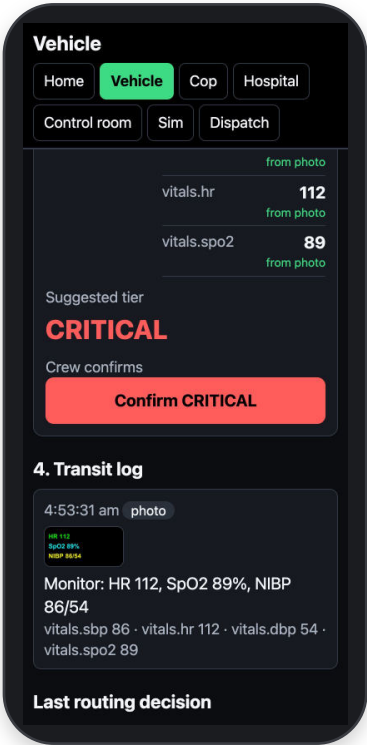
From the crew's words to the constable's phone



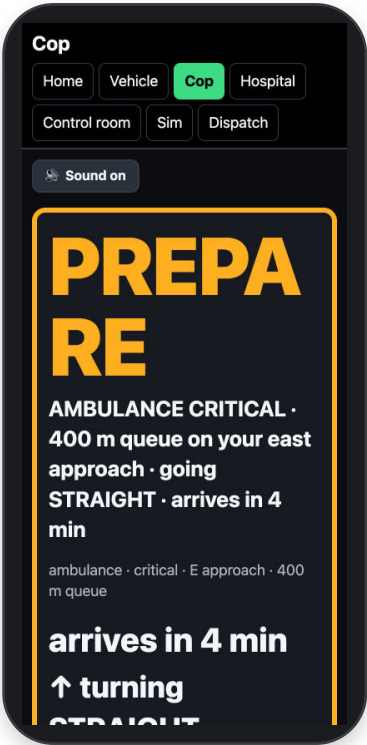
Amber seam: simulated today.

Wireframes

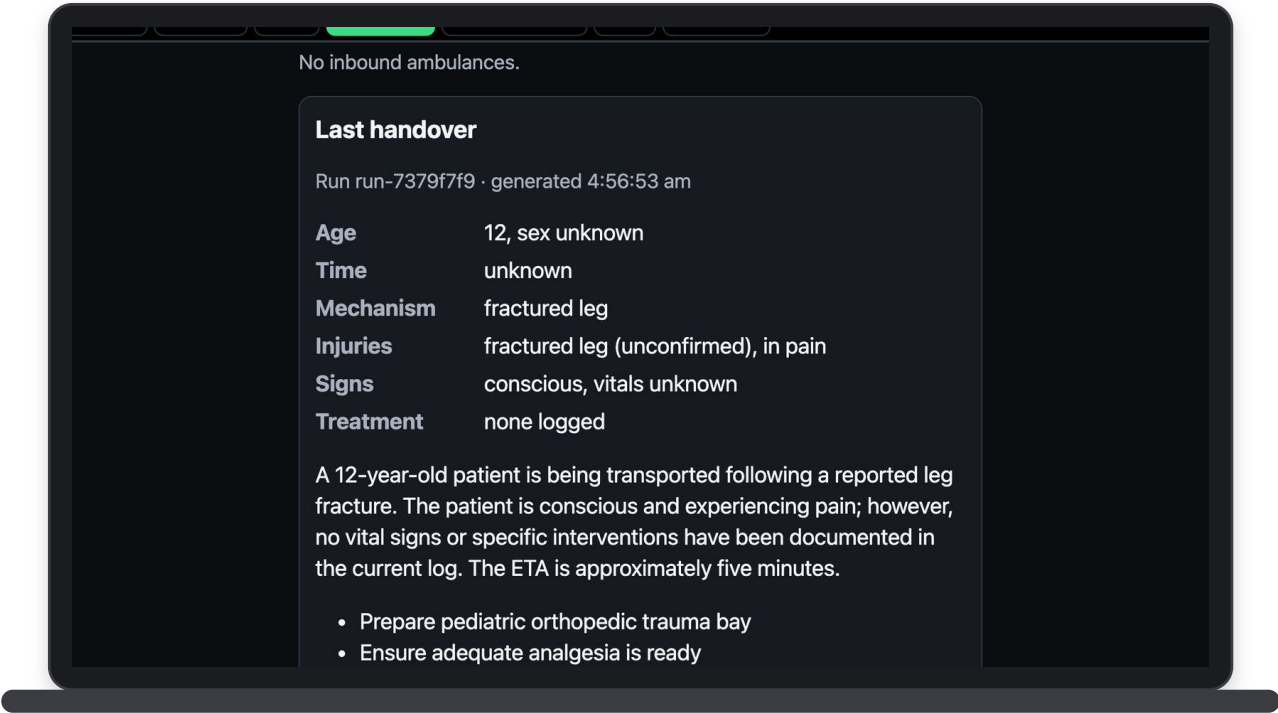
Real screens from the running prototype



Voice triage



PREPARE alert

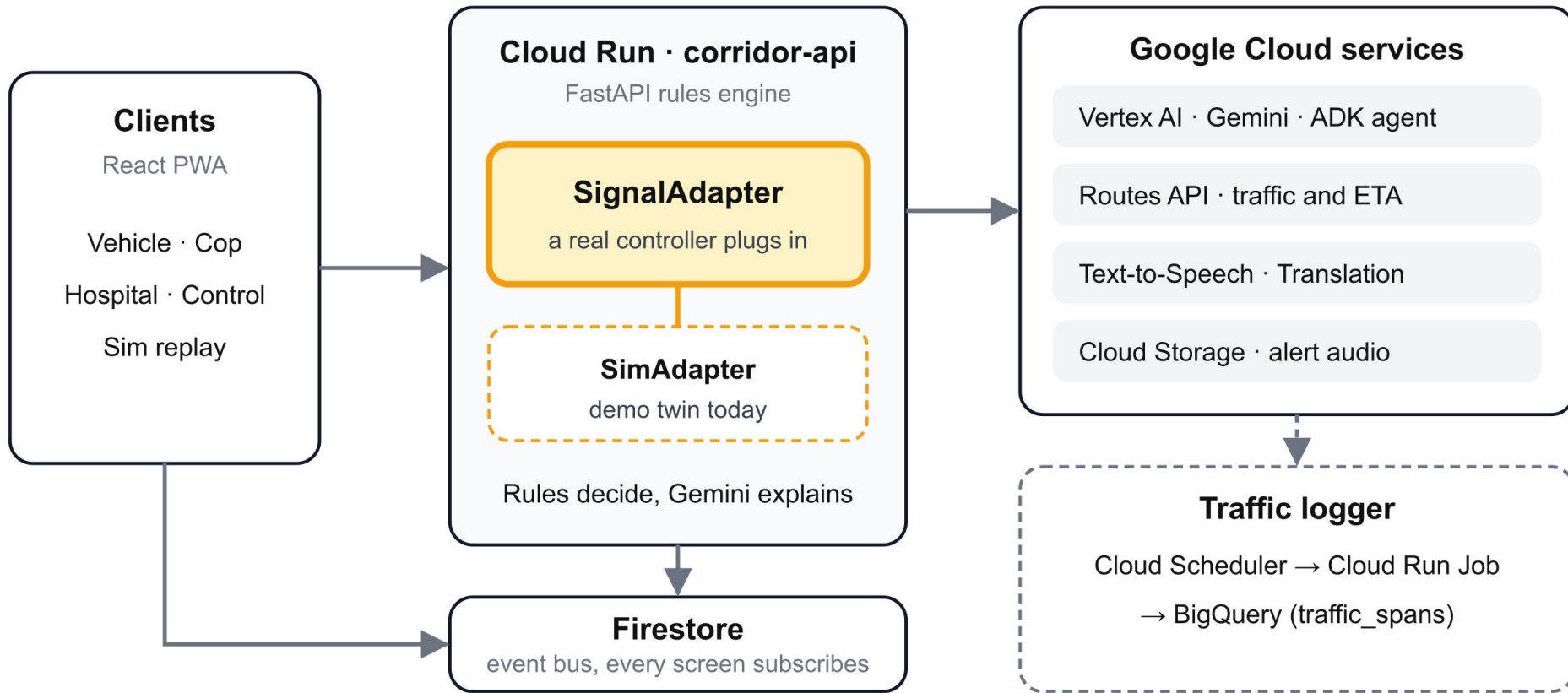


Hospital handover

Live rehearsal captures; synthetic patients.

Architecture

Rules decide; Google Cloud services extract and explain



Rules engine on Cloud Run, Firestore event bus; the amber SignalAdapter seam is where a real controller plugs in.

Technologies (Google Cloud)

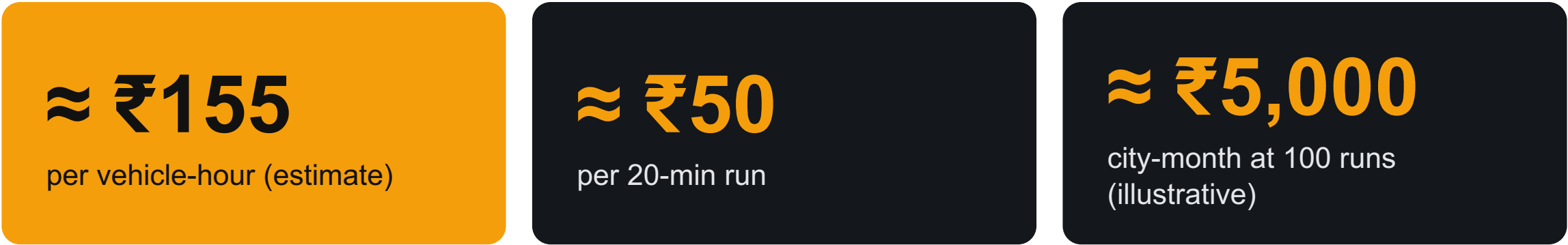
A Google Cloud stack, from model to delivery

● AI & language	Gemini on Vertex AI	Google ADK	Cloud Translation
● Maps & speech	Routes API	Text-to-Speech	Cloud Storage
● Run & data	Cloud Run	Firestore	BigQuery
● Delivery	Firebase Hosting	Cloud Build	Secret Manager

BigQuery ML pipeline in place; peak-hour spans logging on Cloud Scheduler since 6 Oct; model retrained before submission

Estimated implementation cost

About ₹155 per vehicle-hour, almost all of it Routes

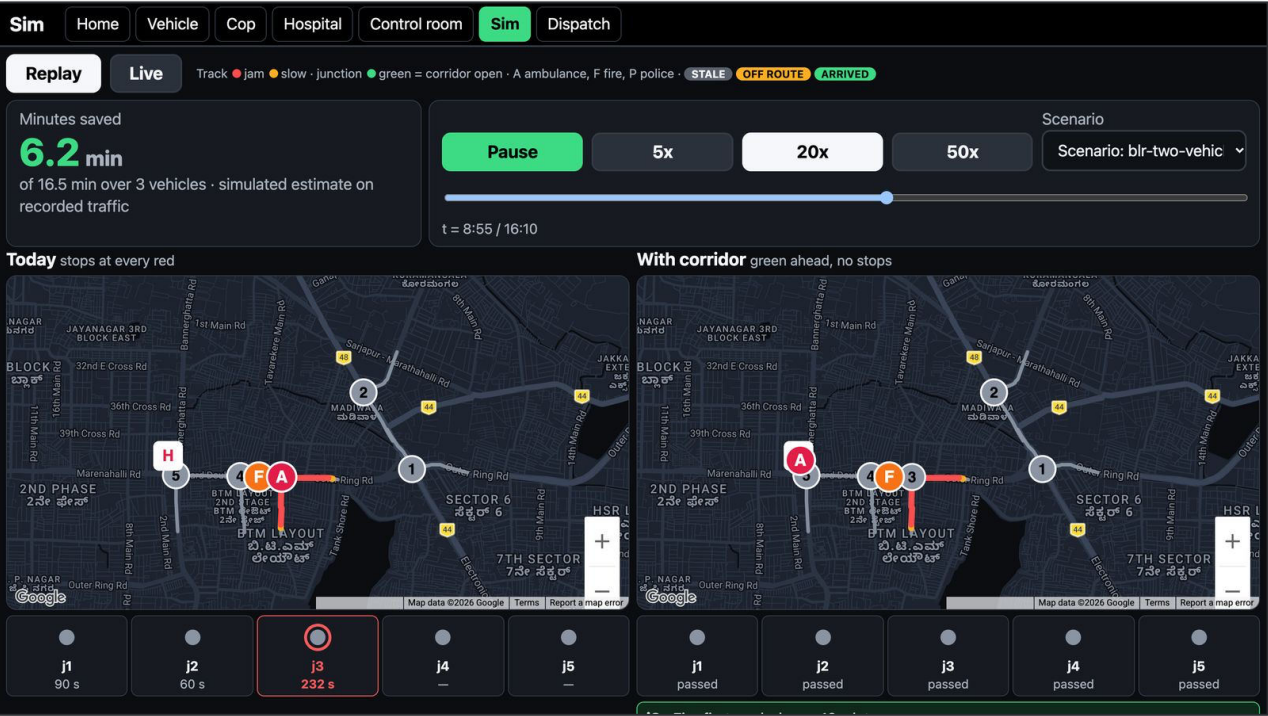


Routes	3/min × 60 × ₹0.85	₹153
Gemini	17 × ₹0.02	₹0.34
Text-to-Speech		₹1.30

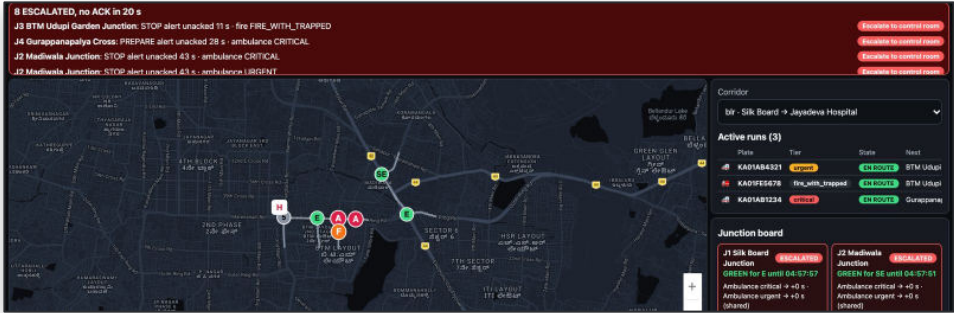
Lever: cache Routes beyond today's 20 s

Snapshots of the prototype

The loop, mid-run, from the live rehearsal



Replay simulator



Control room



STOP alert, spoken

Snapshots of the prototype (routing agent)

Live routing trace: Jayadeva chosen, two alternatives rejected

```
1 required_capabilities(critical) -> cath_lab
2 list_hospitals(blr) -> 3 hospitals
3 check_diversion(Jayadeva) -> accepting
4 check_diversion(Apollo) -> on diversion
5 eta_to(Jayadeva) -> 373 s, 2.6 km
6 Added to baseline: icu
```

Destination Jayadeva, confidence 1.0

Jayadeva Institute of Cardiovascular Sciences · 6:13

recommendation only (scenario run)

- Jayadeva Institute is the most suitable facility as it has both the required cath lab and ICU capabilities, with 4 beds available.
- It was chosen over Apollo Hospital because Apollo is currently on diversion, and Fortis does not have a cath lab.

```
1. called required_capabilities(critical) -> cath_lab
2. called list_hospitals(blr) -> 3 hospitals: blr_jayadeva, blr_apollo_bg,
   blr_fortis_bg
3. called check_diversion(Jayadeva Institute of Cardiovascular Sciences) ->
   accepting
4. called check_diversion(Apollo Hospital Bannerghatta Road) -> on diversion
5. called eta_to(Jayadeva Institute of Cardiovascular Sciences) -> 373 s, 2.6 km
6. capabilities: cath_lab (Required for immediate intervention for suspected
   STEMI/ACS.); icu (Required for patient with cardiogenic shock (hypotension,
   tachycardia) and respiratory distress.). Added to baseline: icu
```

The trace as shown in /vehicle.

Performance report / benchmarking

Synthetic-voice results, scripted-run timings

90 %

field accuracy

100 %

tier accuracy

3.5 s

mean extraction

12 synthetic clips, 11 scored for tier, 1 intervention note; not field recordings. Before the fix: 87 % / 73 %.

17 / 37 ms

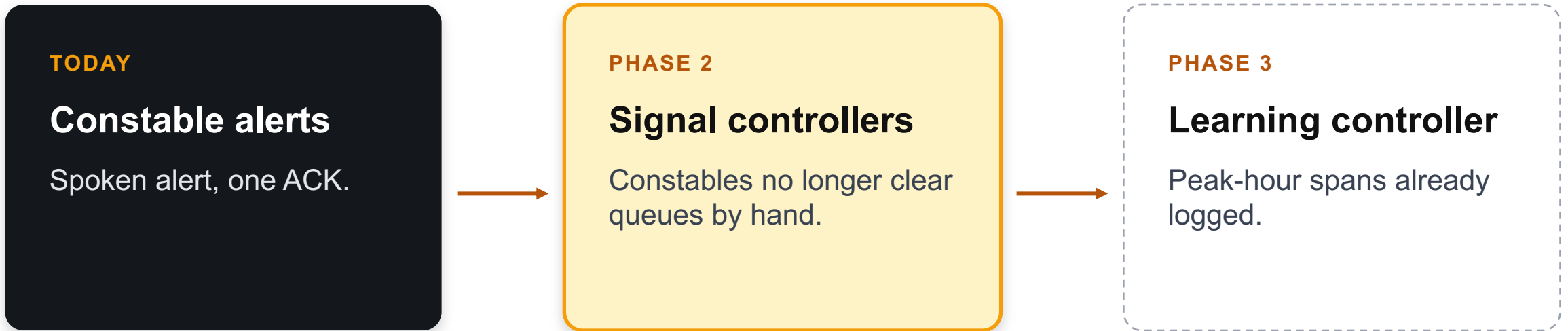
/location p50 / p95, offline

412 Python · **161** web tests

98 % coverage

Additional details / future development

From constable to controller, in three steps



Signals simulated, patients synthetic, minutes saved estimated.

Sustainability

We size the green to the queue, so cross-traffic idling is bounded by the clearance time.

0.25_L

idling fuel avoided per cleared
queue (illustration: 20 vehicles ×
60 s × 0.76 L/h)

2.35_{kg}

CO₂ per litre of petrol (EPA)

SDG 3.8 · 11.2

emergency care access ·
sustainable transport

Idle burn: Argonne National Laboratory (Gaines, Rask, Keller), US DOE, 2014. CO₂: US EPA, GHG Equivalencies Calculator, 2024. The per-queue arithmetic is an illustration, not a measurement or a net saving.

Links

Try it, read the code, watch the demo

GITHUB PUBLIC REPOSITORY



github.com/Nandish3010/ideal-disco

DEMO VIDEO (3 MINUTES)

[link + QR at
submission]

FINAL PRODUCT



green-corridor-2026.web.app

SUBMISSION DECK



green-corridor-2026.web.app/deck.pdf

Warn the constable before the siren arrives.

Try it: green-corridor-2026.web.app

Read it: github.com/Nandish3010/ideal-disco

Built by Nandish